Best-Execution Review

2024-04-01 to 2024-06-14 · Demo Client

Prepared for	Demo Client
Engagement	DEMO-0008
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Executive summary

Demonstration. Synthetic manager, brokers and orders generated for illustration.

This document records the monitoring of execution quality of Demo Client for 2024-04-01 to 2024-06-14 against the best-execution policy. It covers 1,139 orders; 32 could not be measured and are listed. It is prepared for the head of trading and the compliance officer; it documents what the data shows and what has to be explained. It is not a regulatory opinion.

Orders with at least one exception: 444 of 1,139 (39.0%)

An exception is a reason to ask for an explanation, not proof of a breach. Each one is in the order review sheet with columns for the trader's explanation and the compliance review.

Policy limits

Rule	Limit
Fill beyond the order's limit price	tolerance 0 bps
Slippage against the arrival price	60 / 50 / 40 bps (low / normal / high urgency)
Participation in the market volume of the execution interval	50%
Time from decision to arrival at the desk	20 minutes
Commissions and fees	2 bps of notional
Share of notional with one broker	40%
Decision price recorded	required

Source: config.yaml (policy)

Exceptions by rule

Rule	Orders	Share of orders	Share of notional	Most exceptions
limit_breach	25	2.2%	2.1%	Gamma Capital
slippage	83	7.3%	14.0%	Gamma Capital
participation	171	15.0%	28.4%	Delta Direct
delay	68	6.0%	8.4%	Alpha Securities
commission	235	20.6%	22.6%	Gamma Capital
decision_price	15	1.3%	1.2%	Alpha Securities

Source: tables/policy_summary.csv

Methodology

Each limit is tested on every order from the fills and the market bars. Slippage is measured against the arrival price and compared with the limit for the order's urgency; participation is the executed quantity over the market volume of the execution interval; delay is the time between the decision and the arrival at the desk; a limit breach is a fill at a price worse than the order's limit price; commissions are commissions and fees over the notional. The limits are those agreed at kickoff and recorded in the configuration of the run.

Results

1. Broker selection and concentration

The largest broker is Alpha Securities with 39% of the notional (limit 40%).

Broker	Orders	Notional	Shortfall	vs arrival	vs VWAP	Explicit	Reversion	Model	Alpha	Low	High	Result
Gamma Capital	235	22%	34.9	33.2	+25.0	2.5	15.7	16.0	+17.2	9.3	25.6	worse than the model
Delta Direct	159	13%	21.0	19.5	+20.6	0.8	18.8	14.9	+4.6	2.1	7.0	worse than the model
Alpha Securities	418	39%	19.0	13.7	+14.7	1.6	19.2	16.2	-2.5	-8.4	2.8	not significant
Beta Markets	327	26%	17.5	10.3	+14.8	1.3	15.0	15.8	-5.6	-11.6	0.4	not significant

Source: tables/by_broker.csv

2. Venues

Venue	Fills	Notional	Slippage	Reversion
NASDAQ	3,129	24%	19.0	18.9
DARK-X	3,050	24%	16.4	18.8
NYSE	2,538	21%	17.7	16.8
IEX	1,915	16%	17.5	18.6
ARCA	1,831	15%	18.8	18.5

Source: tables/by_venue.csv

3. Exceptions by broker

Broker	commission	decision_price	delay	limit_breach	participation	slippage
Alpha Securities	0	6	24	0	19	30
Beta Markets	0	4	19	0	1	21
Delta Direct	0	3	10	0	89	0
Gamma Capital	235	2	15	25	62	32

Source: tables/policy_exceptions.csv

4. Limit-price breaches

Order	Date	Symbol	Broker	Algorithm	Notional	Detail
O00575	2024-05-06	KPR	Gamma Capital	TWAP	18,941,253	2 fill(s) beyond the limit price 220.44
O01023	2024-06-04	NVX	Gamma Capital	MARKET	12,436,202	1 fill(s) beyond the limit price 189.56
O00531	2024-05-02	EMR	Gamma Capital	TWAP	8,996,893	2 fill(s) beyond the limit price 107.93
O00301	2024-04-18	BRX	Gamma Capital	POV	7,430,118	4 fill(s) beyond the limit price 238.24
O00760	2024-05-17	IRN	Gamma Capital	MARKET	4,860,457	1 fill(s) beyond the limit price 121.21
O00997	2024-06-03	NVX	Gamma Capital	MARKET	4,665,840	1 fill(s) beyond the limit price 188.43
O00334	2024-04-22	NVX	Gamma Capital	POV	3,715,321	5 fill(s) beyond the limit price 194.83
O00074	2024-04-04	PLM	Gamma Capital	TWAP	3,317,428	2 fill(s) beyond the limit price 182.30
O00364	2024-04-23	MRD	Gamma Capital	POV	3,209,380	3 fill(s) beyond the limit price 167.17
O00751	2024-05-17	QTZ	Gamma Capital	MARKET	2,864,319	1 fill(s) beyond the limit price 144.94
O01061	2024-06-06	MRD	Gamma Capital	TWAP	2,063,486	5 fill(s) beyond the limit price 164.76
O00316	2024-04-19	PLM	Gamma Capital	TWAP	1,849,820	2 fill(s) beyond the limit price 178.33
O00257	2024-04-16	ALD	Gamma Capital	TWAP	1,533,640	2 fill(s) beyond the limit price 219.06
O00713	2024-05-15	IRN	Gamma Capital	TWAP	1,338,161	5 fill(s) beyond the limit price 118.62
O00270	2024-04-17	PLM	Gamma Capital	POV	1,217,890	6 fill(s) beyond the limit price 179.89
O00390	2024-04-24	EMR	Gamma Capital	TWAP	1,100,843	5 fill(s) beyond the limit price 108.13
O00998	2024-06-03	EMR	Gamma Capital	TWAP	1,037,101	6 fill(s) beyond the limit price 103.05
O00131	2024-04-09	ORB	Gamma Capital	POV	954,561	2 fill(s) beyond the limit price 227.50
O00655	2024-05-10	FRG	Gamma Capital	TWAP	916,620	11 fill(s) beyond the limit price 83.98
O00528	2024-05-02	NVX	Gamma Capital	POV	556,084	1 fill(s) beyond the limit price 191.60
O00139	2024-04-09	FRG	Gamma Capital	MARKET	466,574	1 fill(s) beyond the limit price 89.55
O00228	2024-04-15	DLT	Gamma Capital	POV	270,524	7 fill(s) beyond the limit price 28.11
O00419	2024-04-25	GLX	Gamma Capital	POV	242,271	2 fill(s) beyond the limit price 28.84
O00937	2024-05-29	CVN	Gamma Capital	TWAP	236,270	9 fill(s) beyond the limit price 94.20
O00490	2024-04-30	TRX	Gamma Capital	TWAP	216,924	4 fill(s) beyond the limit price 53.92

Source: tables/policy_exceptions.csv

5. Largest exceptions

Order	Date	Symbol	Side	Broker	Algorithm	Rule	Detail
O00658	2024-05-10	KPR	Buy	Gamma Capital	POV	commission	explicit costs 2.7 bps of the notional
O01176	2024-06-13	KPR	Buy	Alpha Securities	VWAP	slippage	slippage against arrival 55 bps, normal urgency
O00846	2024-05-23	NVX	Sell	Alpha Securities	VWAP	participation	95% of the volume of the execution interval
O00289	2024-04-18	EMR	Buy	Alpha Securities	VWAP	participation	53% of the volume of the execution interval
O00644	2024-05-09	RVN	Sell	Gamma Capital	MARKET	commission	explicit costs 2.7 bps of the notional
O00886	2024-05-24	NVX	Sell	Delta Direct	DMA	participation	737% of the volume of the execution interval
O00851	2024-05-23	JVL	Sell	Gamma Capital	MARKET	participation	636% of the volume of the execution interval
O01128	2024-06-11	EMR	Sell	Alpha Securities	VWAP	participation	82% of the volume of the execution interval
O01069	2024-06-06	LMN	Sell	Beta Markets	TWAP	slippage	slippage against arrival 62 bps, low urgency
O01131	2024-06-11	BRX	Sell	Alpha Securities	IS	participation	131% of the volume of the execution interval
O00101	2024-04-05	BRX	Buy	Alpha Securities	IS	participation	65% of the volume of the execution interval
O00981	2024-05-31	PLM	Sell	Alpha Securities	VWAP	delay	36 minutes between decision and arrival
O00717	2024-05-15	JVL	Sell	Delta Direct	DMA	delay	33 minutes between decision and arrival
O00244	2024-04-15	KPR	Buy	Delta Direct	DMA	participation	167% of the volume of the execution interval
O01002	2024-06-03	STL	Sell	Beta Markets	POV	slippage	slippage against arrival 51 bps, high urgency
O00534	2024-05-02	KPR	Buy	Gamma Capital	TWAP	slippage	slippage against arrival 66 bps, normal urgency
O01075	2024-06-07	RVN	Buy	Gamma Capital	MARKET	participation	300% of the volume of the execution interval
O01000	2024-06-03	RVN	Buy	Gamma Capital	MARKET	commission	explicit costs 2.7 bps of the notional
O00989	2024-05-31	JVL	Buy	Alpha Securities	IS	slippage	slippage against arrival 83 bps, low urgency
O00575	2024-05-06	KPR	Buy	Gamma Capital	TWAP	commission	explicit costs 2.7 bps of the notional
O00650	2024-05-09	IRN	Buy	Gamma Capital	MARKET	slippage	slippage against arrival 59 bps, high urgency
O00816	2024-05-22	RVN	Buy	Delta Direct	DMA	participation	884% of the volume of the execution interval
O00141	2024-04-09	KPR	Sell	Gamma Capital	MARKET	participation	448% of the volume of the execution interval
O00847	2024-05-23	PLM	Sell	Beta Markets	VWAP	slippage	slippage against arrival 63 bps, low urgency
O00563	2024-05-06	JVL	Buy	Alpha Securities	IS	participation	96% of the volume of the execution interval
O00049	2024-04-03	EMR	Sell	Beta Markets	VWAP	delay	43 minutes between decision and arrival
O00805	2024-05-21	BRX	Buy	Alpha Securities	IS	slippage	slippage against arrival 46 bps, high urgency
O00152	2024-04-10	EMR	Sell	Delta Direct	DMA	participation	137% of the volume of the execution interval
O00491	2024-04-30	EMR	Buy	Gamma Capital	MARKET	participation	136% of the volume of the execution interval
O01114	2024-06-10	PLM	Buy	Delta Direct	DMA	participation	189% of the volume of the execution interval

Source: tables/policy_exceptions.csv

6. Review and sign-off

Exceptions are reviewed in the order review sheet (columns: the trader's explanation, the compliance review, the status). The review is complete when no exception is Open.

Head of trading
Name, signature, date

Compliance officer
Name, signature, date

Limitations and assumptions

- The price data is five-minute bars, so very small differences here mean little.
- The policy limits are internal thresholds agreed at kickoff; they are not regulatory limits.
- An exception can have a good reason (a deliberate decision to trade urgently, a thin market); the review exists to record it.

Appendix

Data quality

Audit trail

Run ID	`tc_20261111T180027Z_660e69`
Run started / finished (UTC)	2026-11-11T18:00:27+00:00
Analyst	info@proquantanalytics.com
Code version (git)	`2703b230a84f (uncommitted changes) · core 0.1.0`
Config SHA-256	`d3e18f911785c174`
Input: orders	`1171 rows · sha256 385479b70739727c`
Input: fills	`12737 rows · sha256 9f474f09f7fed8bf`
Input: bars	`85566 rows · sha256 944f1f24550488b9`
Environment	`python 3.13.5 · pandas 2.2.3 · numpy 2.1.3`

Source: audit.json in the run folder

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